

Zhengzhong Jin

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RESEARCH INTERESTS

Cryptography and related topics in Theoretical Computer Science

EXPERIENCE

Northeastern University <i>Assistant Professor</i>	Aug 2024~
Northeastern University <i>Khoury Senior Research Fellow</i>	Feb 2024 ~ Aug 2024
Massachusetts Institute of Technology <i>Postdoctoral Associate, Supervisor: Vinod Vaikuntanathan</i>	Sept 2022 ~ Jan 2024
University of California, Berkeley <i>Visiting Student Researcher, Host: Sanjam Garg</i>	2021~2022

EDUCATION

Ph. D. in Computer Science , <i>Johns Hopkins University</i> Advisors: Abhishek Jain and Xin Li Thesis: Non-interactive Proofs: What Assumptions are Sufficient?	2017~2022
B.S. in Information and Computing Science , <i>Fudan University</i> School of Mathematical Sciences	2013~2017

AWARDS

Best Paper Award, EUROCRYPT 2021

Best Student Paper Award, ACNS 2019

PROGRAM COMMITTEE

Theory of Cryptography Conference (TCC) 2023
Public Key Cryptography (PKC) 2024
International Cryptology Conference (CRYPTO) 2024
Theory of Cryptography Conference (TCC) 2024

PUBLICATION

- Non-Interactive Zero-Knowledge from LPN and MQ
Quang Dao, Aayush Jain, Zhengzhong Jin
CRYPTO 2024

- SNARGs Under LWE via Propositional Proofs
Zhengzhong Jin, Yael Tauman Kalai, Alex Lombardi, Vinod Vaikuntanathan
STOC 2024
- Scalable Multiparty Garbling
Gabrielle Beck, Aarushi Goel, Aditya Hegde, Abhishek Jain, Zhengzhong Jin, Gabriel Kaptchuk
CCS 2023
- Correlation Intractability and SNARGs from Sub-exponential DDH
Arka Rai Choudhuri, Sanjam Garg, Abhishek Jain, Zhengzhong Jin, Jiaheng Zhang
CRYPTO 2023
- A Note on Non-Interactive Zero-Knowledge Proofs for NP from CDH
Geoffroy Couteau, Abhishek Jain, Zhengzhong Jin, Willy Quach
CRYPTO 2023
- Credibility in Private Set Membership
Sanjam Garg, Mohammad Hajibadi, Abhishek Jain, Zhengzhong Jin, Omkant Pandey, Sina Shiehian
PKC 2023
- Linear Insertion Deletion Codes in the High-Noise and High-Rate Regimes
Kuan Cheng, Zhengzhong Jin, Xin Li, Zhide Wei, Yu Zheng
ICALP 2023
- Succinct Zero Knowledge for Floating Point Computations
Sanjam Garg, Abhishek Jain, Zhengzhong Jin, Yinyu Zhang
CCS 2022
- Indistinguishability Obfuscation via Mathematical Proofs of Equivalence
Abhishek Jain, Zhengzhong Jin
FOCS 2022
- Pre-Constrained Encryption
Prabhanjan Ananth, Abhishek Jain, Zhengzhong Jin, Giulio Malavolta
ITCS 2022
- Compact and Flexible KEM From Ideal Lattice
Zhengzhong Jin, Shiyu Shen, Yunlei Zhao
IEEE TIT 2022
- SNARGs for P from LWE
Arka Rai Choudhuri, Abhishek Jain, Zhengzhong Jin
FOCS 2021, Invited to the SIAM Journal of Computing Special Issue
- Non-Interactive Batch Arguments for NP from Standard Assumptions
Arka Rai Choudhuri, Abhishek Jain, Zhengzhong Jin
CRYPTO 2021
- Non-Interactive Zero Knowledge from Sub-exponential DDH
Abhishek Jain, Zhengzhong Jin
EUROCRYPT 2021, Best Paper Award, Invited to the Journal of Cryptology

- Unbounded Multi-Party Computation from Learning with Errors
Prabhanjan Ananth, Abhishek Jain, Zhengzhong Jin, Giulio Malavolta
EUROCRYPT 2021
- Streaming and Small Space Approximation Algorithms for Edit Distance and Longest Common Subsequence
Kuan Cheng, Alireza Farhadi, MohammadTaghi Hajiaghayi, Zhengzhong Jin, Xin Li, Aviad Rubinstein, Saeed Seddighin, Yu Zheng
ICALP 2021
- Multi-key Fully-Homomorphic Encryption in the Plain Model
Prabhanjan Ananth, Abhishek Jain, Zhengzhong Jin, Giulio Malavolta
TCC 2020
- Statistical Zaps and New Oblivious Transfer Protocols
Vipul Goyal, Abhishek Jain, Zhengzhong Jin, Giulio Malavolta
EUROCRYPT 2020
- Function Hiding Functional Encryption for Pattern Matching
James Bartusek, Brent Carmer, Abhishek Jain, Zhengzhong Jin, Tancrede Lepoint, Fermi Ma, Tal Malkin, Alex J. Malozemoff, Mariana Raykova
ASIACRYPT 2019
- Edit Errors with Block Transpositions: Deterministic Document Exchange Protocols and Almost Optimal Binary Codes
Kuan Cheng, Zhengzhong Jin, Xin Li, Ke Wu
ICALP 2019
- Deterministic Document Exchange Protocols, and Almost Optimal Binary Codes for Edit Errors
Kuan Cheng, Zhengzhong Jin, Xin Li, Ke Wu
FOCS 2018, Journal of the ACM
- Generic and Practical Key Establishment from Lattice
Zhengzhong Jin, Yunlei Zhao
ACNS 2019, Best Student Paper Award

INVITED TALKS

SNARGs Under LWE via Propositional Proofs

NTT Research, March 2024.

iO via Mathematical Proofs of Equivalence

Oxford-Warwick Complexity Meetings, Oct 2022.

63rd IEEE Symposium on Foundations of Computer Science, (FOCS 2022).

Charles River Crypto Day, Dec 2022.

CMU CyLab Crypto Seminar, Feb 2023.

Simons Institute, Proof Complexity and Meta-Mathematics, March 2023.

SNARGs for P from LWE

Charles River Crypto Day, July 2021.
DC Area Crypto Day, Nov 2021.
University of California, Los Angeles, Dec 2021.
62nd IEEE Symposium on Foundations of Computer Science, (FOCS 2021).
Chinese Academy of Sciences, Nov 2022.

Non-Interactive Zero Knowledge from Sub-exponential DDH

UC Berkeley Crypto Seminar (virtual), March 2021.
JHU Crypto reading group (virtual), April 2021.
MIT CIS Seminar (virtual), April 2021.
Cornell Crypto Seminar (virtual), April 2021.
IIT Madras (virtual), July 2021.
40th Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT 2021)
ENSL/CWI/RHUL joint online seminar (virtual), May 2022.

Multi-key Fully Homomorphic Encryption in the Plain Model

Theory of Cryptography Conference, (TCC 2020).

Statistical Zaps and New Oblivious Transfer Protocols

39th Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT 2020)

Block Edit Errors with Transpositions:

Deterministic Document Exchange Protocols and Almost Optimal Binary Codes

46th International Colloquium on Automata, Languages and Programming (ICALP 2019)

Generic and Practical Key Establishment from Lattice

17th International Conference on Applied Cryptography and Network Security
(ACNS 2019)

Deterministic Document Exchange Protocols, and

Almost Optimal Binary Codes for Edit Errors

Johns Hopkins University, Theory Seminar, September 2018.
59th Annual IEEE Symposium on Foundations of Computer Science (FOCS 2018)